

Wrong Models in Ecology I

Neutral Theory: Demographic Stochasticity & Ecological Equivalence

Jacopo Grilli, Bengaluru 14/1/2025

A week ago



Statistical ecology

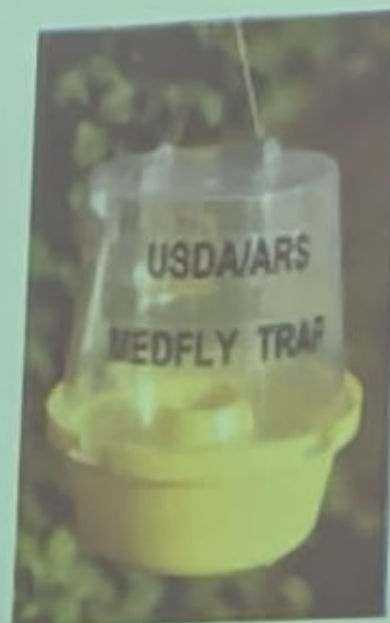
Predict species "fitness" (presence, abundance, reproduction, other measurements...) from environment & own traits

Ecological data

Controlled studies



Natural observation



Data matrix

		Sample					Descriptor			
		a	b	c	d	e	I	II	III	IV
Species	1									
	2									
	3									
	4	Abundance data					Species data			
	5									
	6									
	7									
Factor	X									
	Y	Environment data								
	Z									
	W									

Tends to be noisy and sparse

Today and Tomorrow

Species	Sample				
	a	b	c	d	e
	1				
	2				
	3				
	4				
	5				
	6				
	7				

Abundance data

Tends to be noisy and sparse

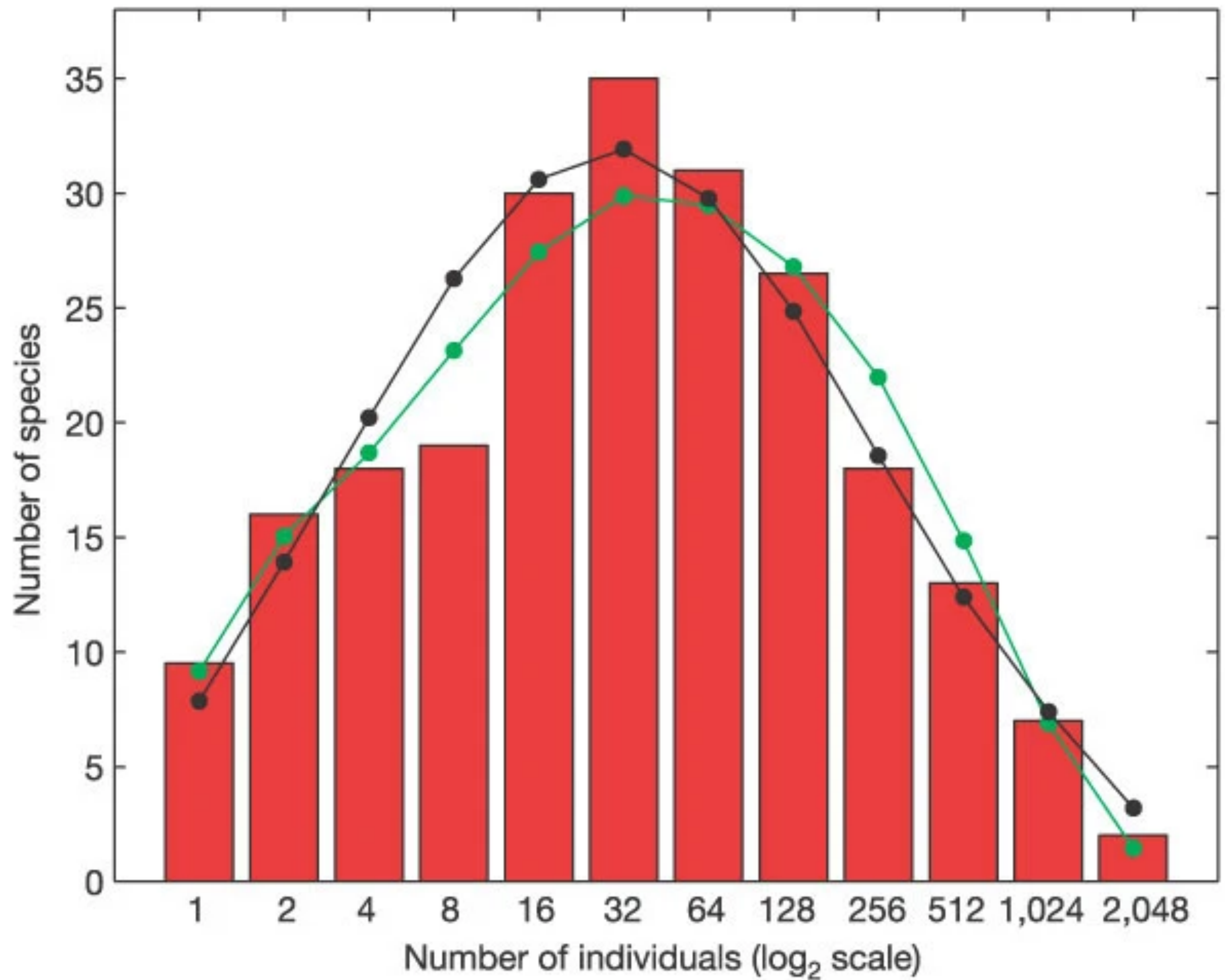
	a	
Species	1	
	2	
	3	Ab
	4	
	5	
	6	
	7	

A very basic phenomenology

There are many different “things” (species/
types)

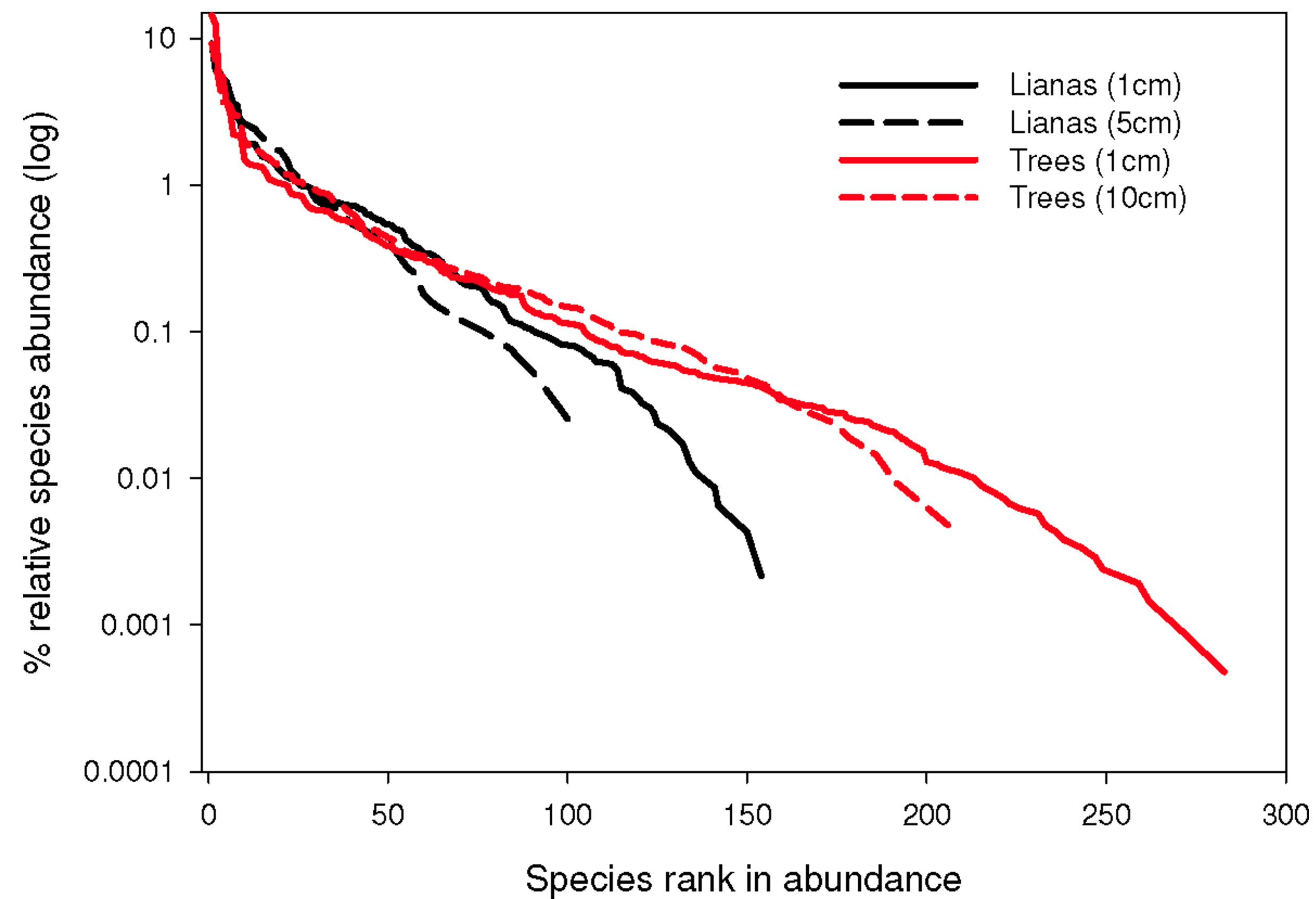
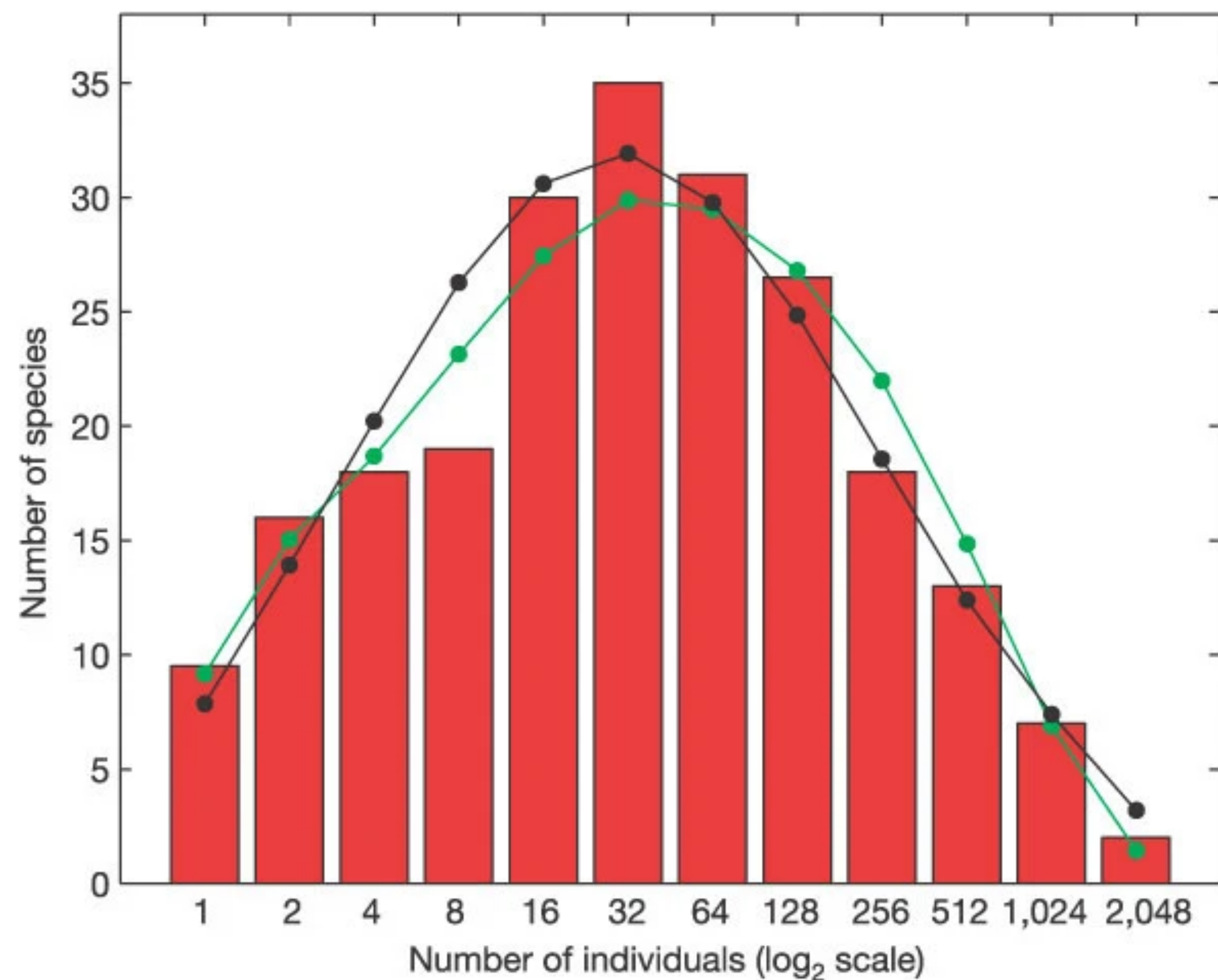
There are “common” and “rare” species

Making the observation more quantitative

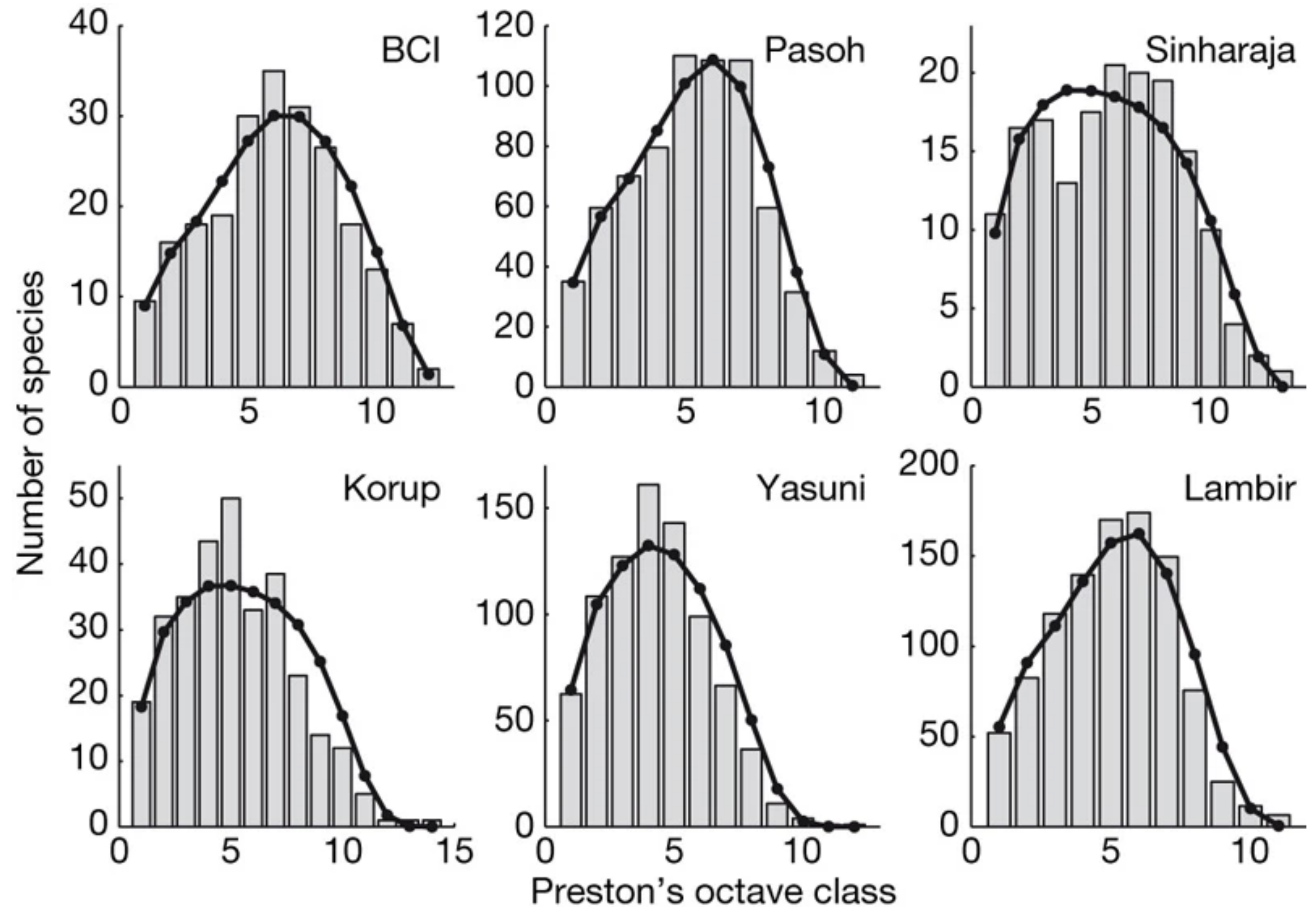


[Volkov et al., 2003]

Same observation in a different form

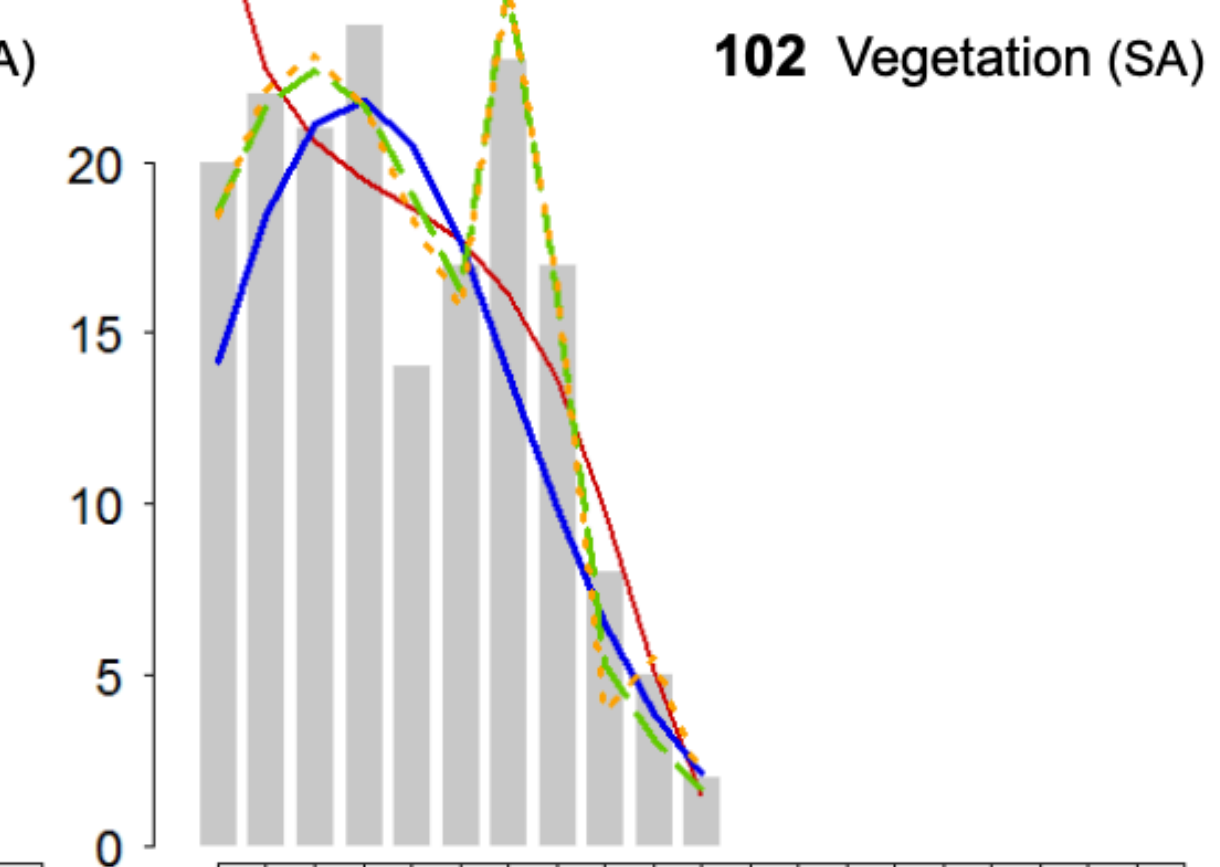
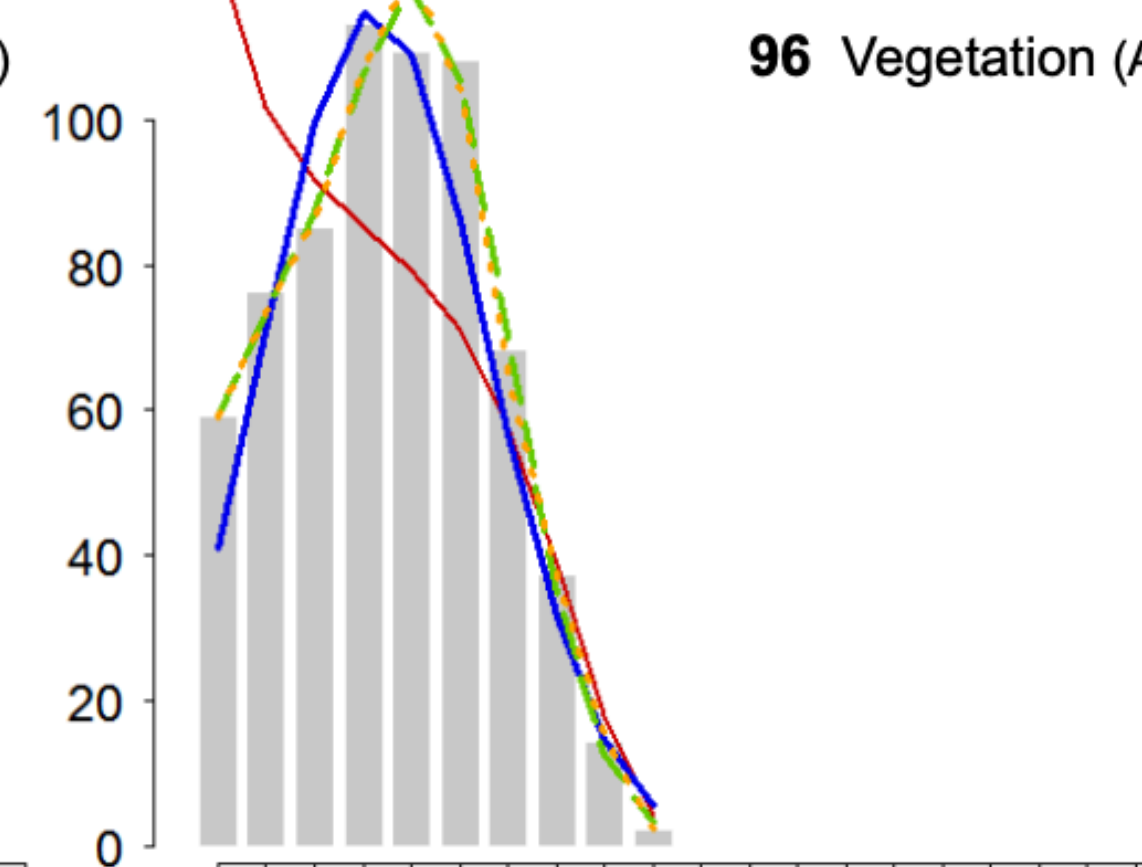
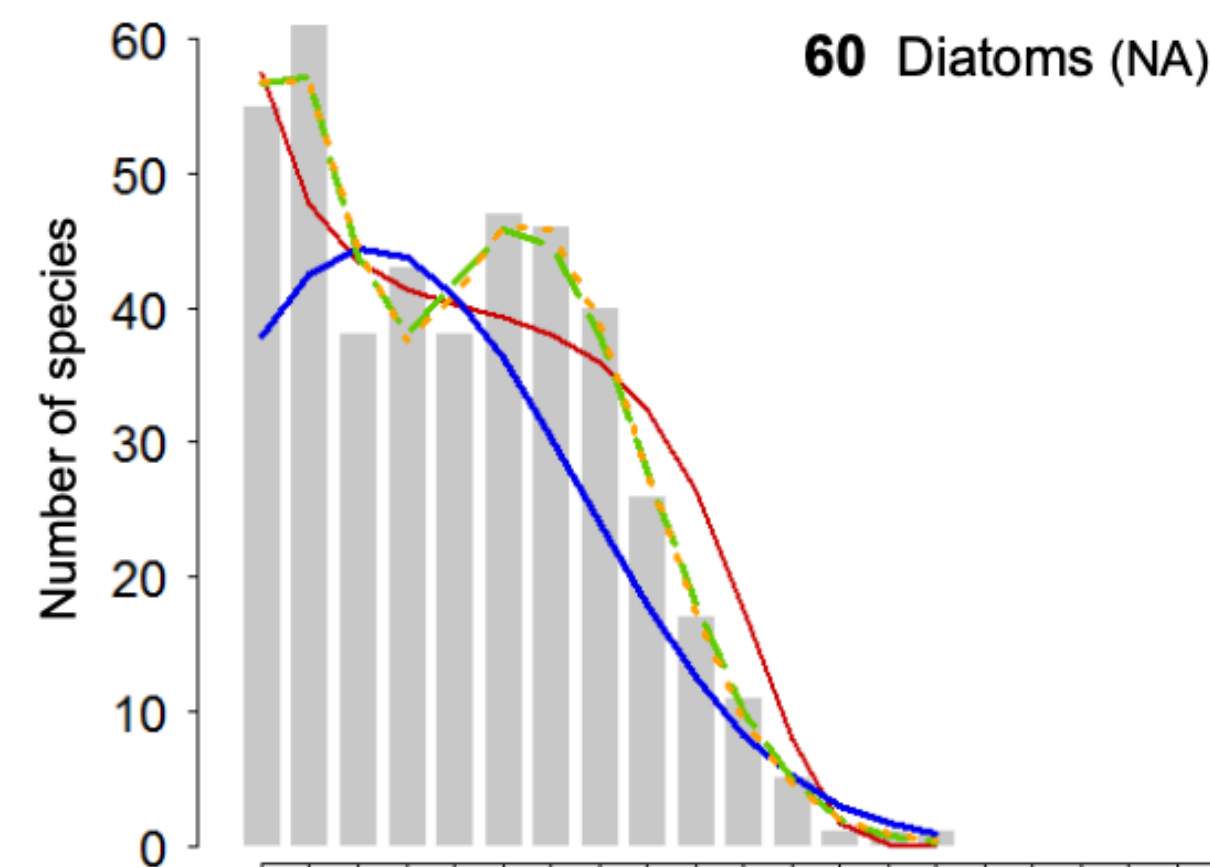
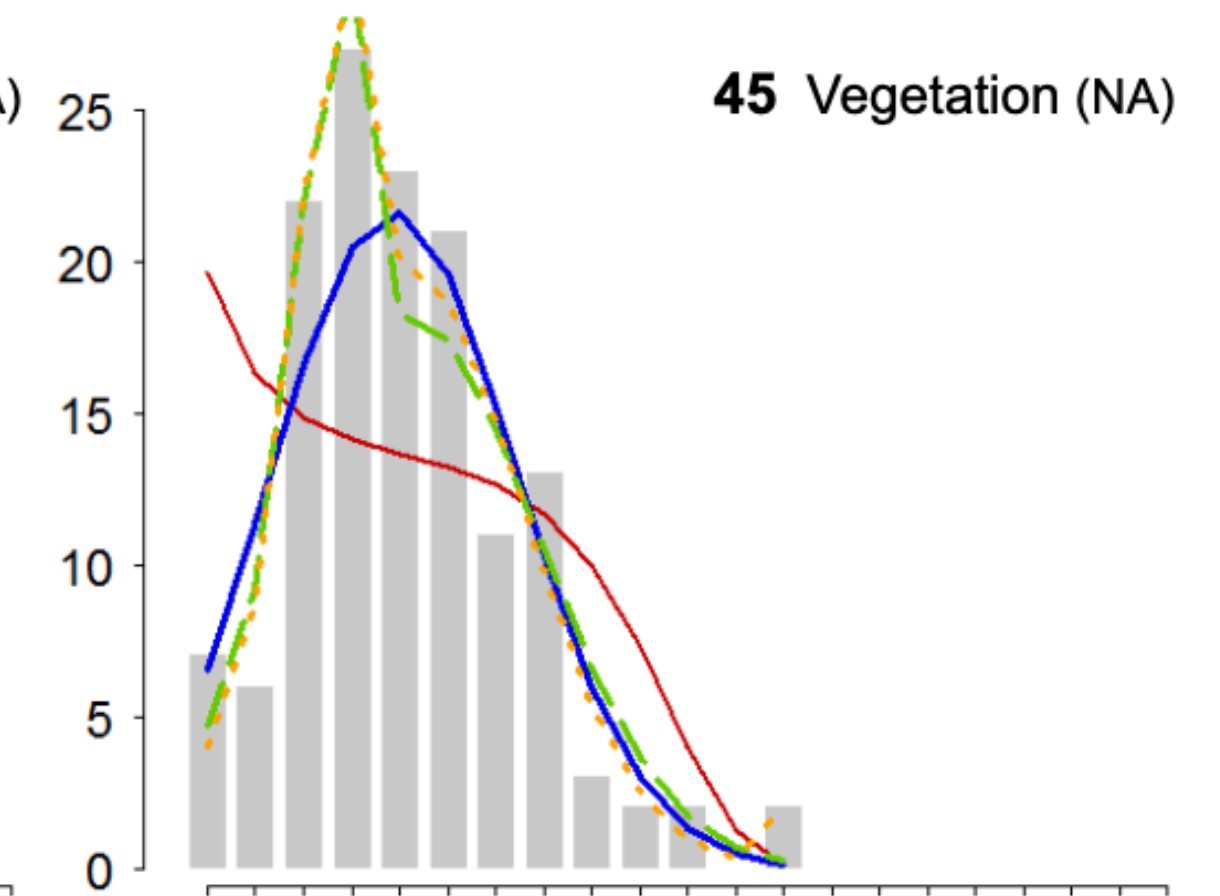
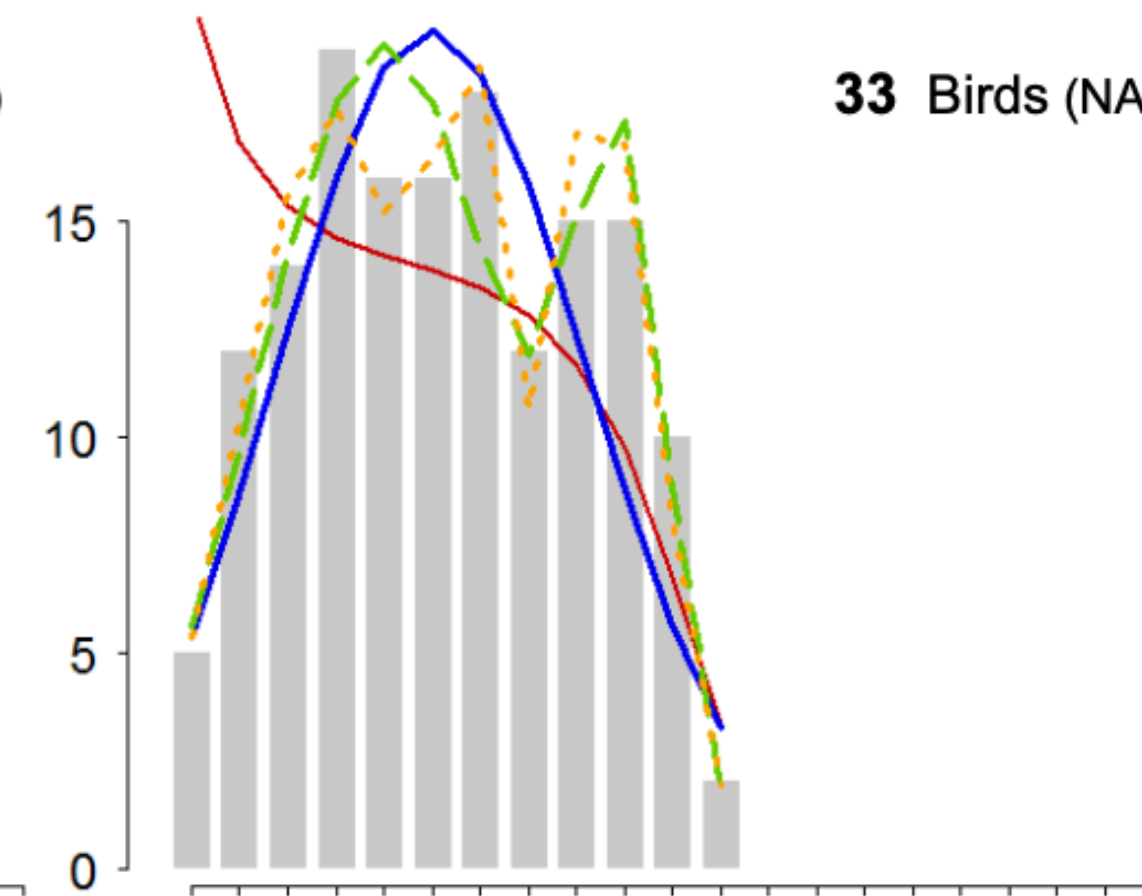
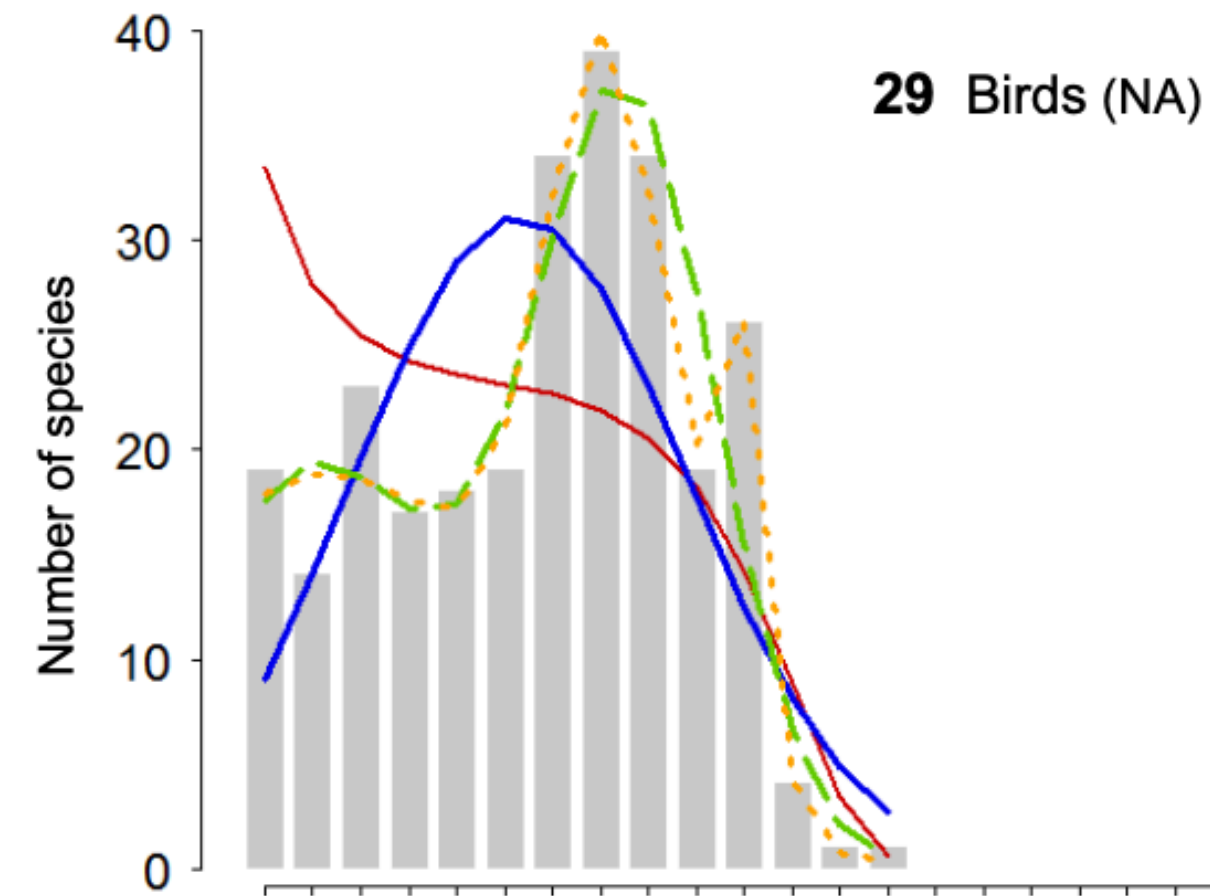
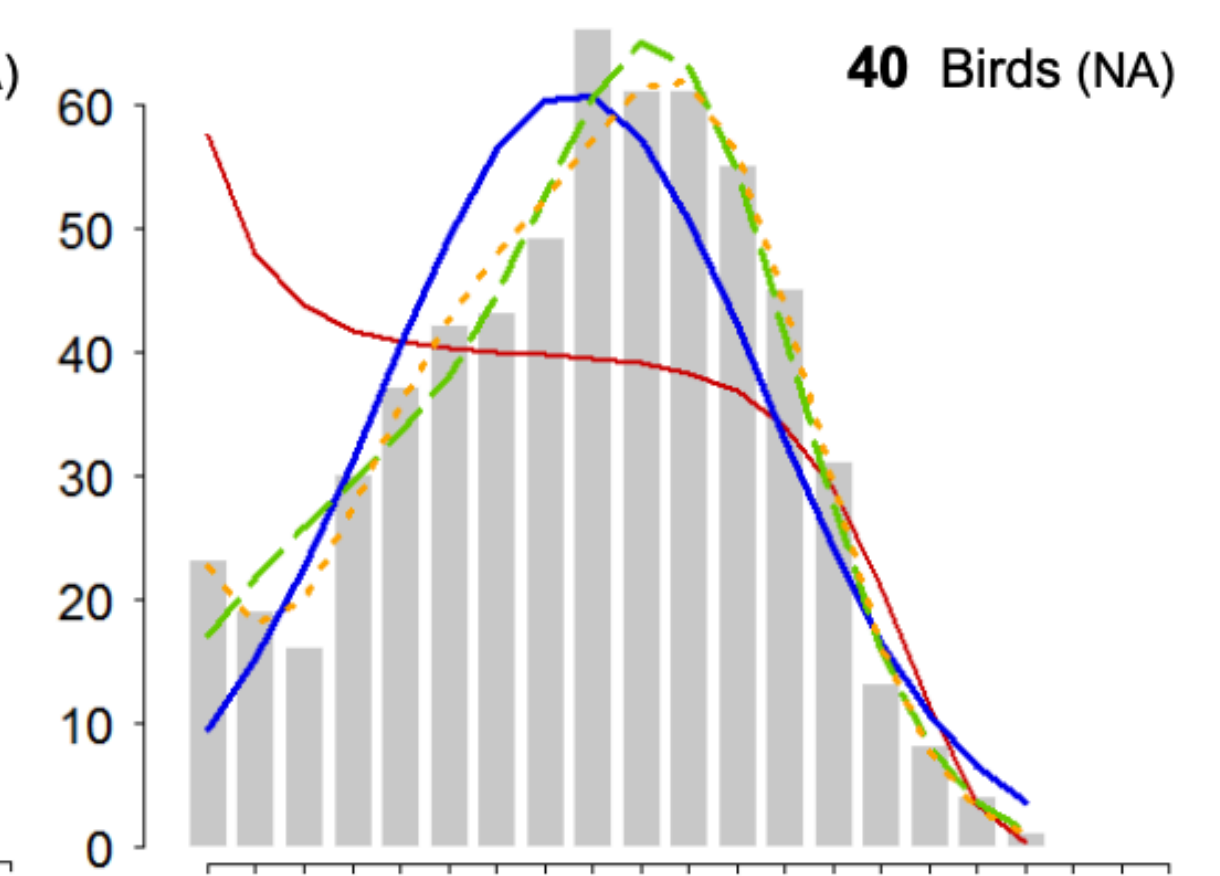
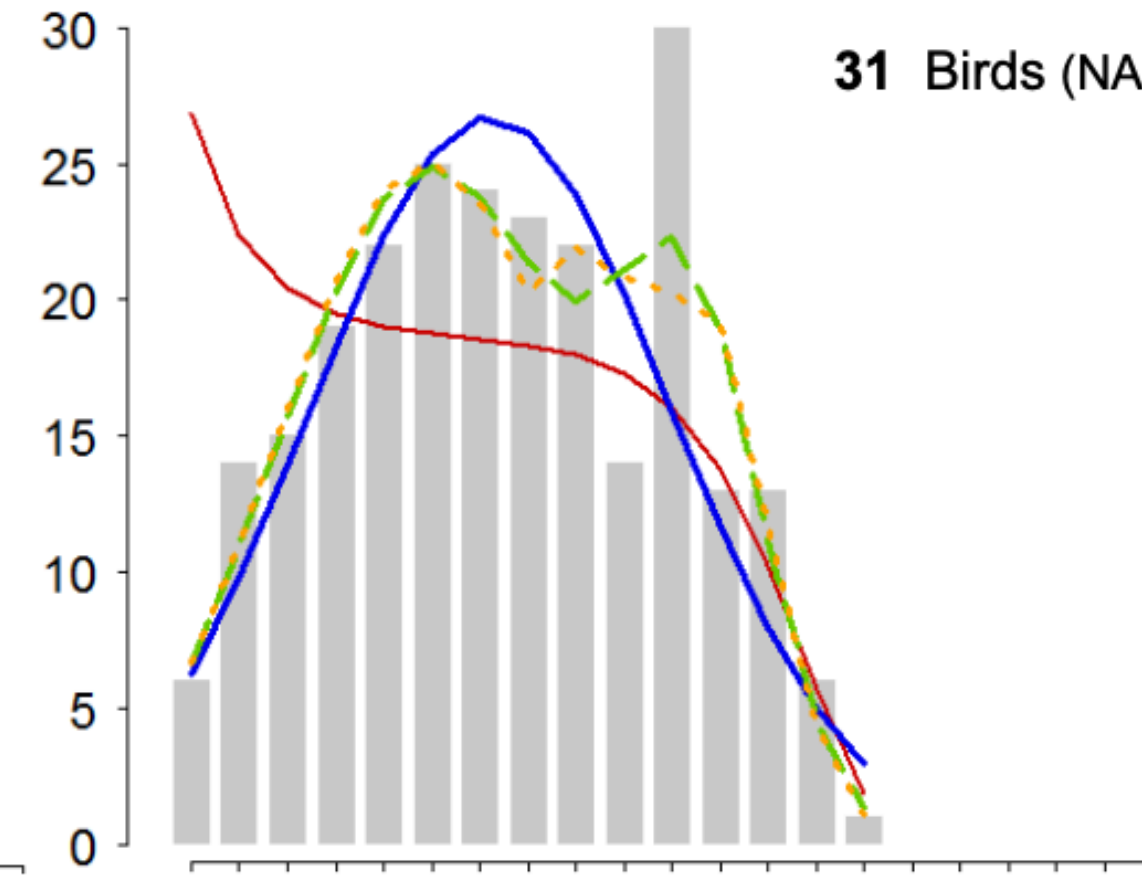
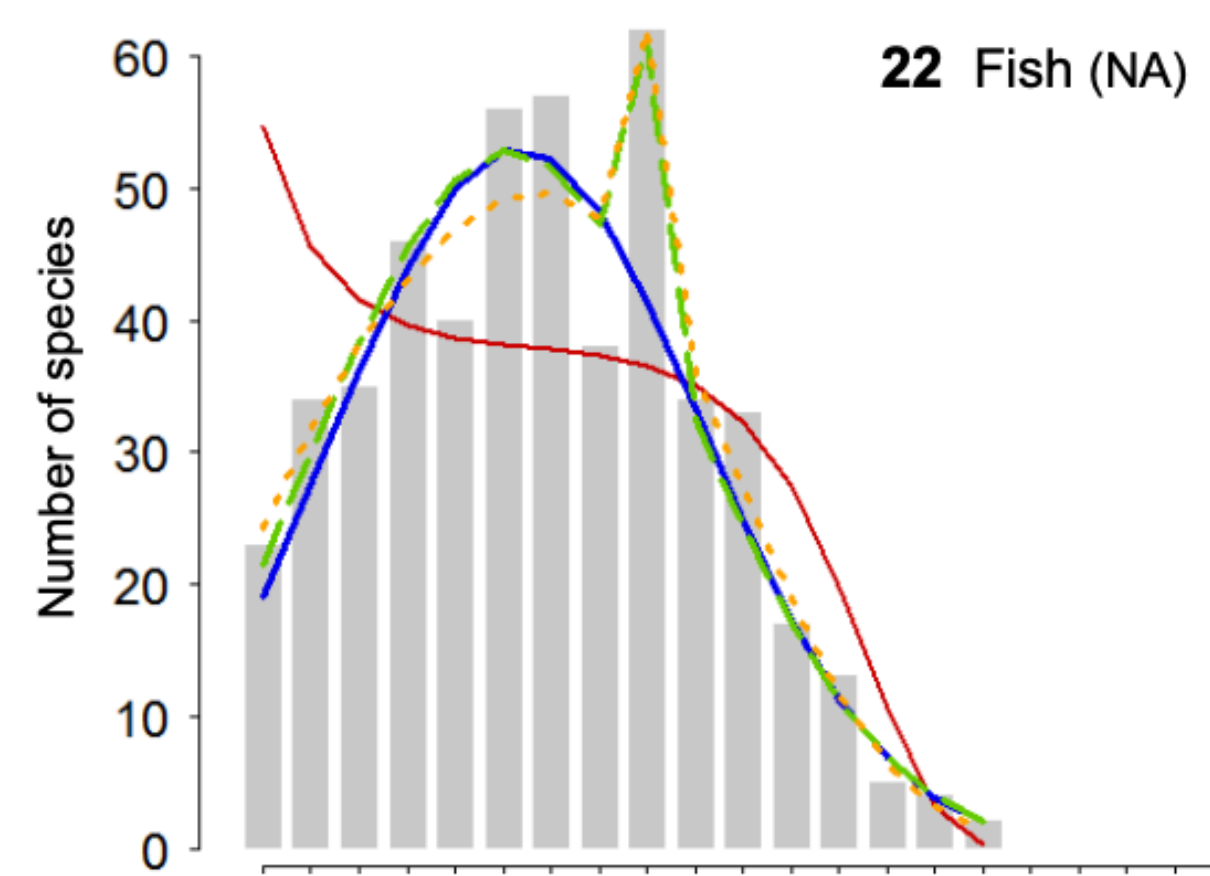


Across tropical forests



[Volkov et al., 2007]

not
just
trees

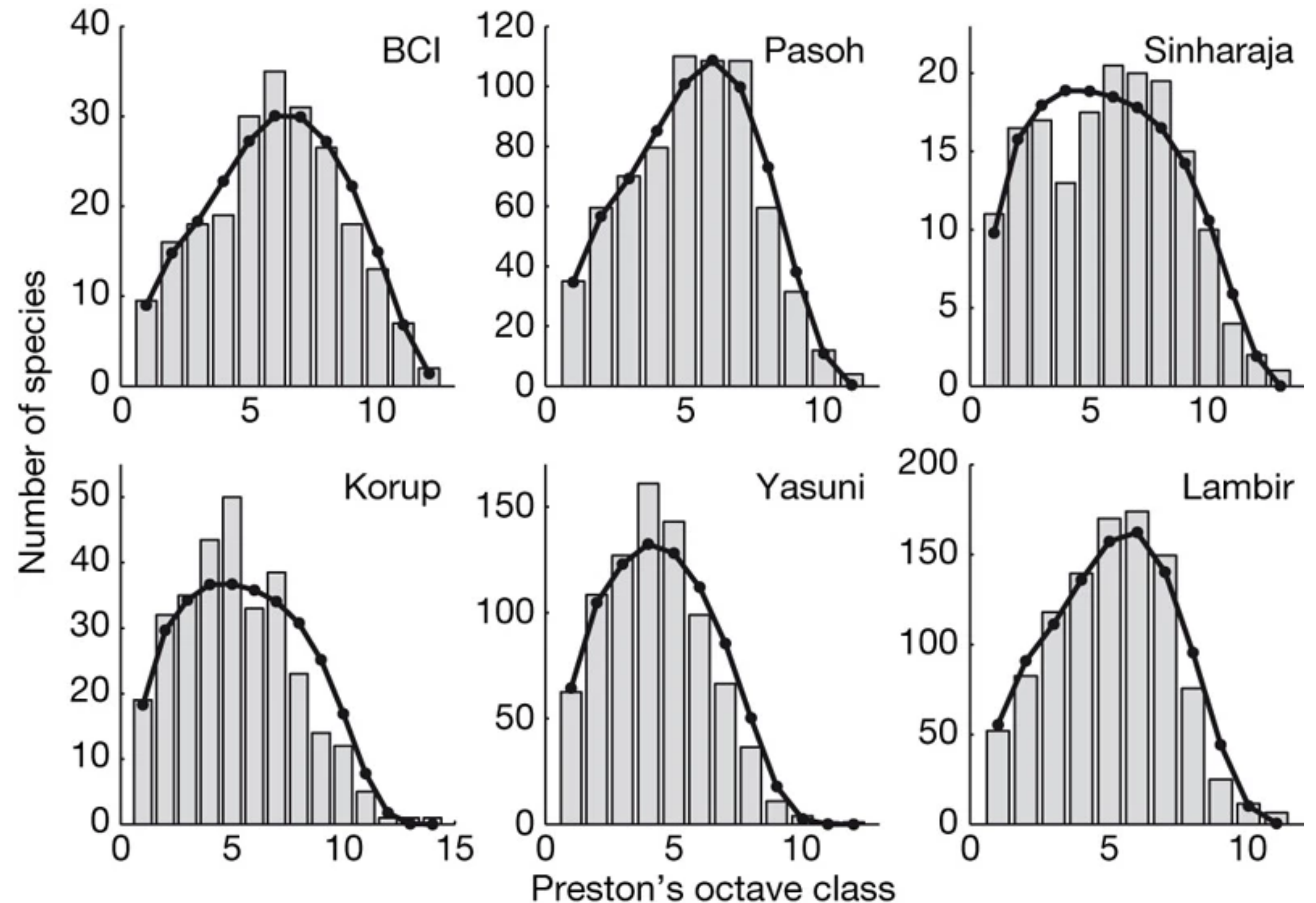


What we should expect “by chance”?
what is a minimal model?

<- board

NT reproduces species abundace distributions

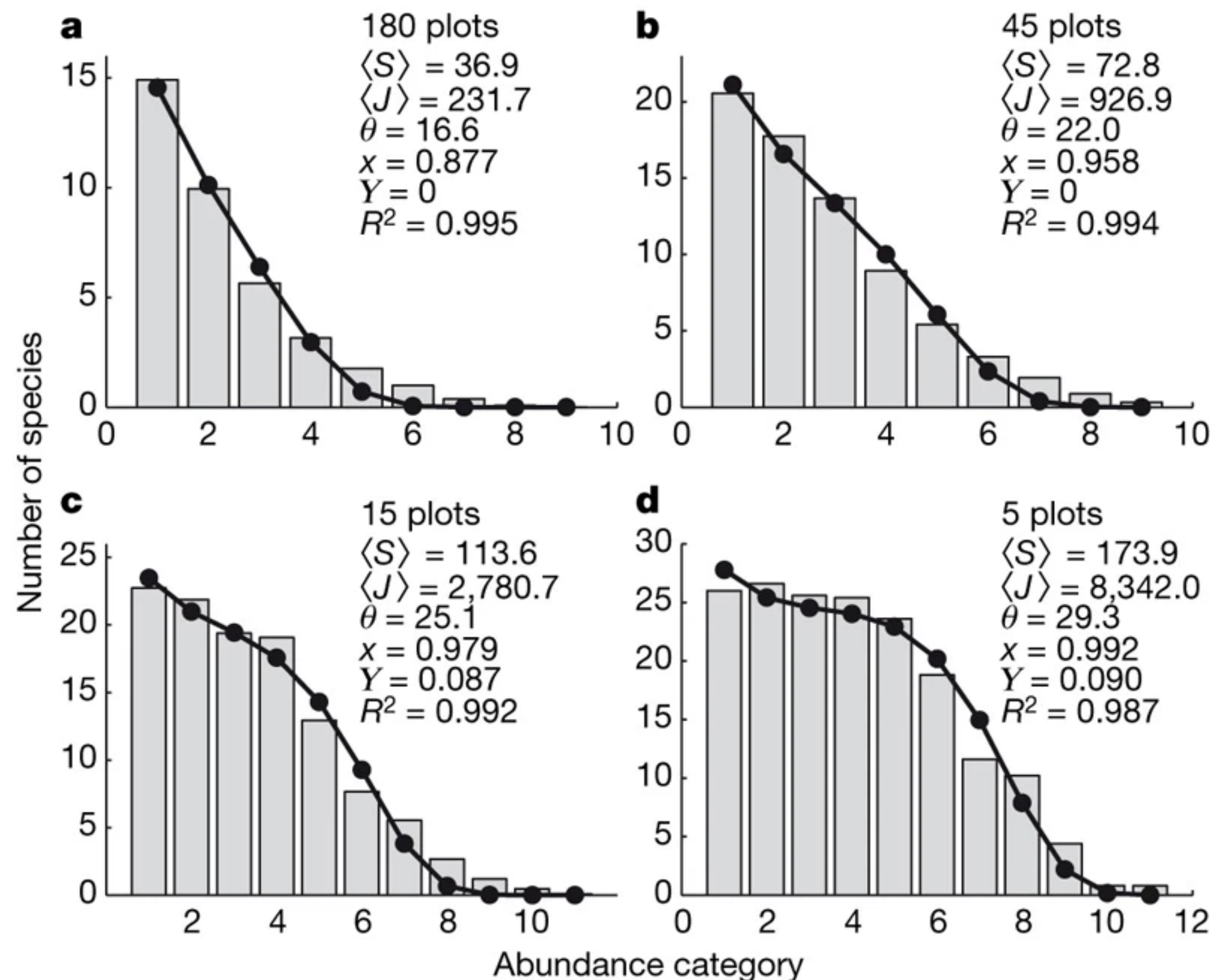
**tropical
forests**



[Volkov et al., 2007]

NT reproduces species abundance distributions

coral
reefs



[Volkov et al., 2007]

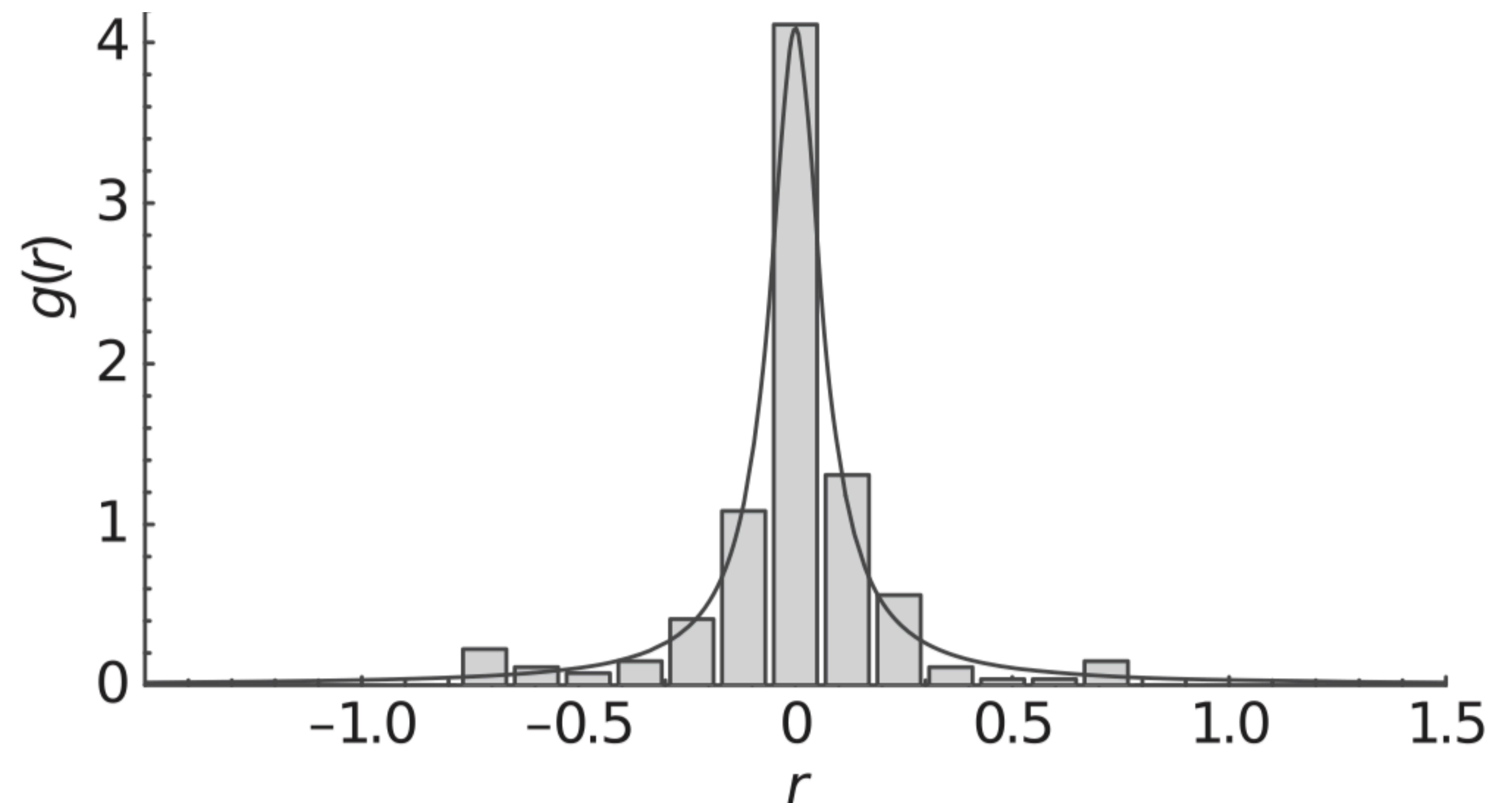
What else can you do with neutral theory?

Dynamics

$$\begin{aligned}\mathcal{P}(\lambda, t) &= \langle \delta(\lambda - x/x_0) \rangle \\ &= \int_0^\infty dx_0 \int_0^\infty dx P(x, t|x_0, 0) P_0(x_0) \delta(\lambda - x/x_0)\end{aligned}$$

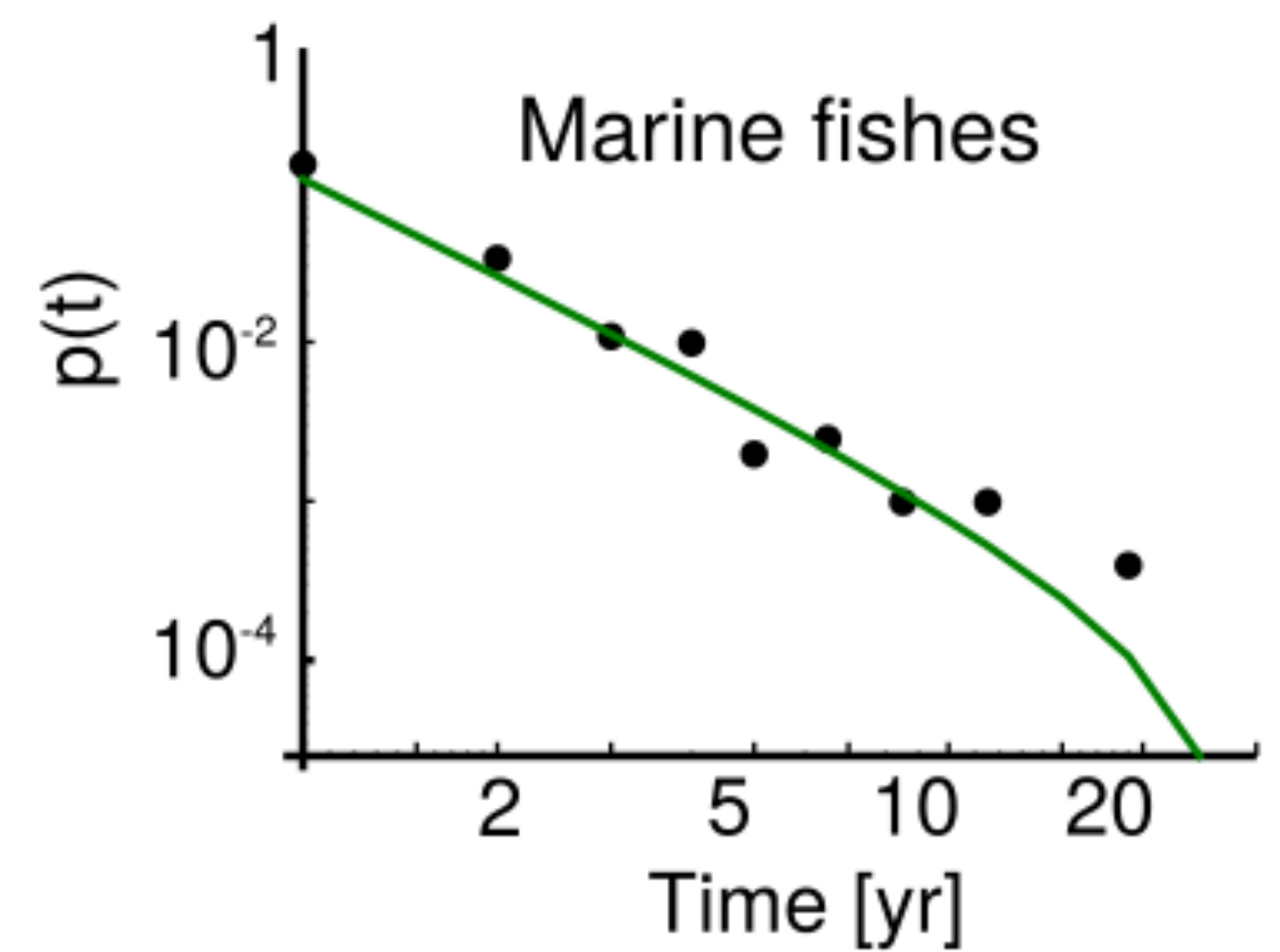
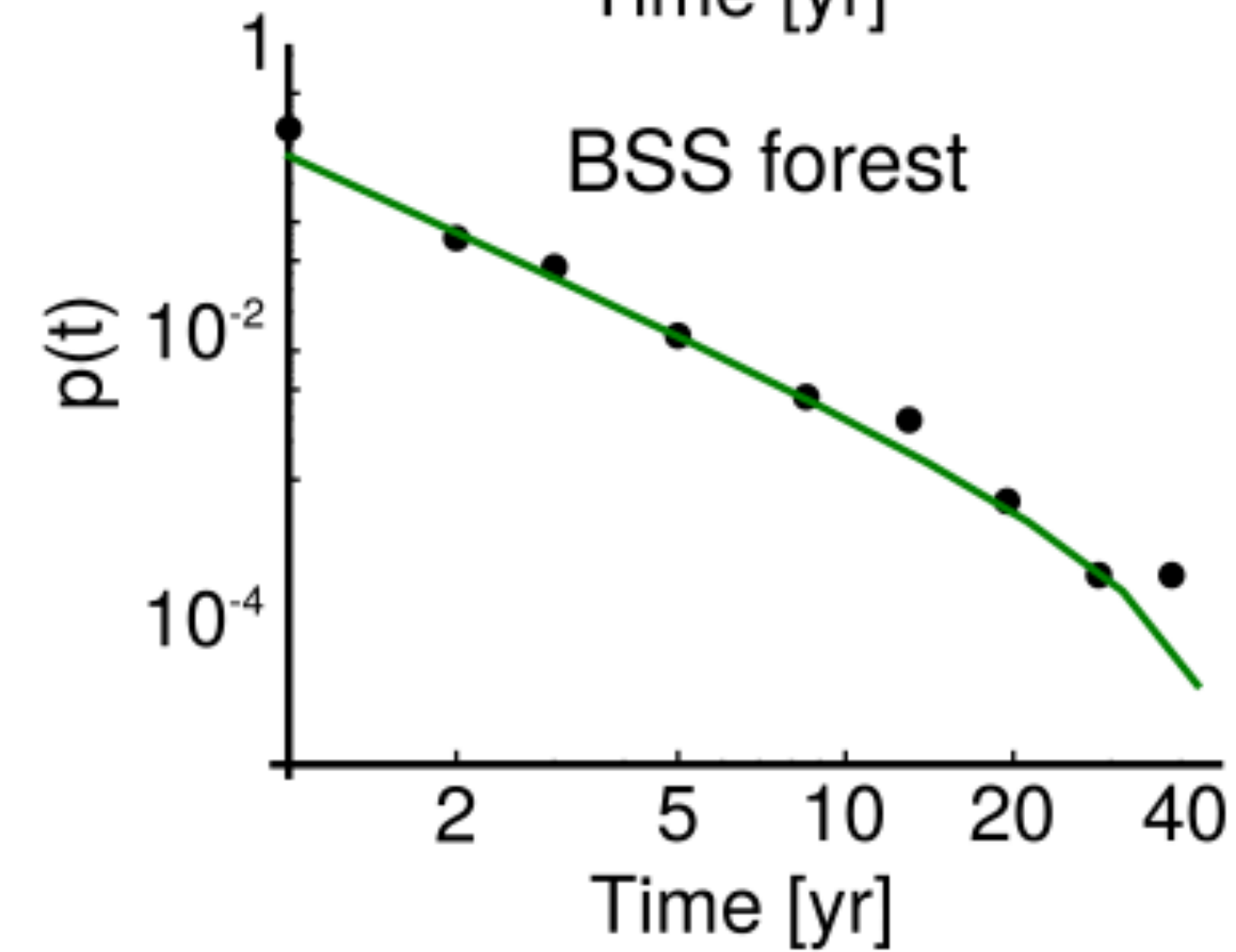
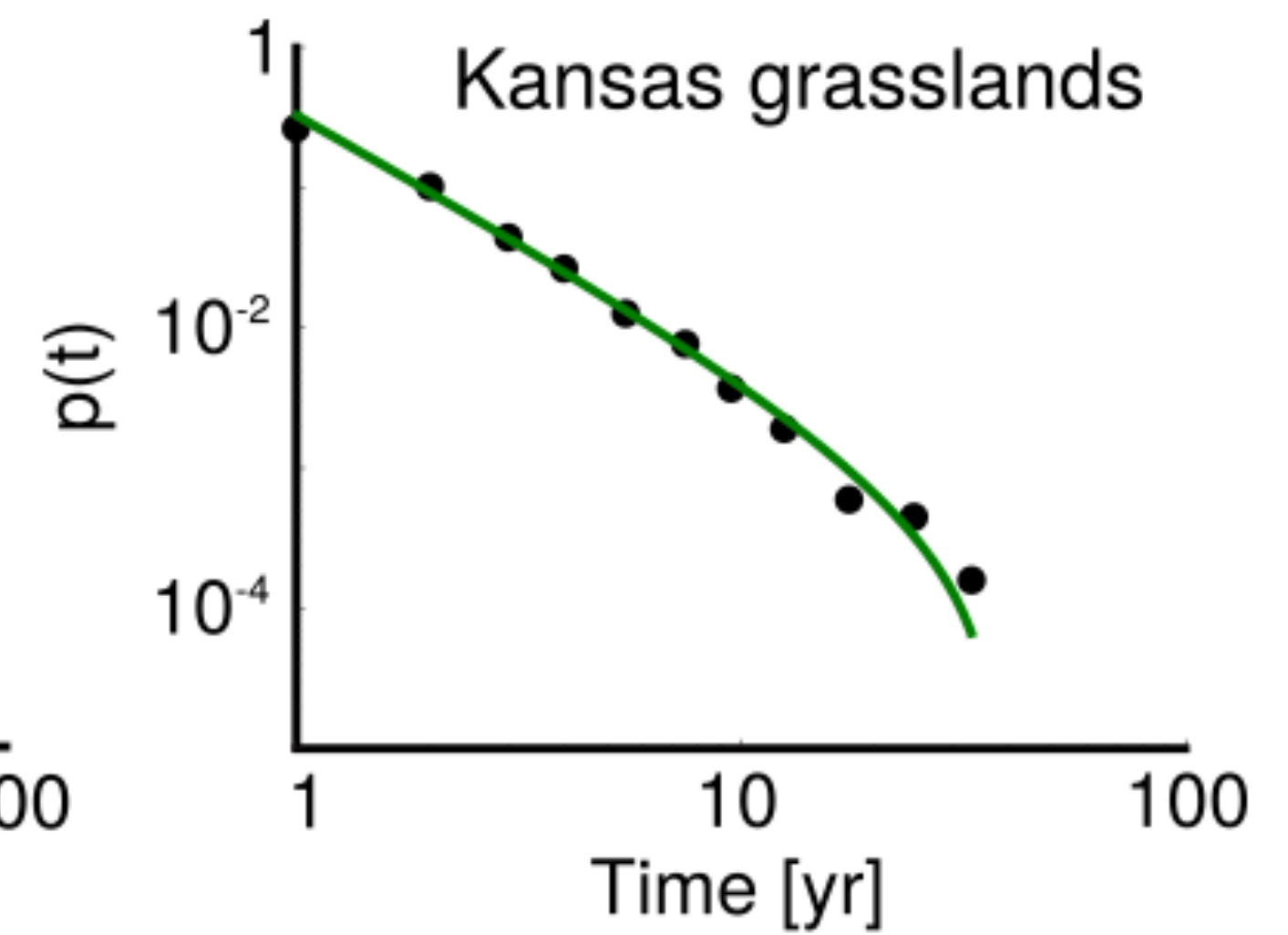
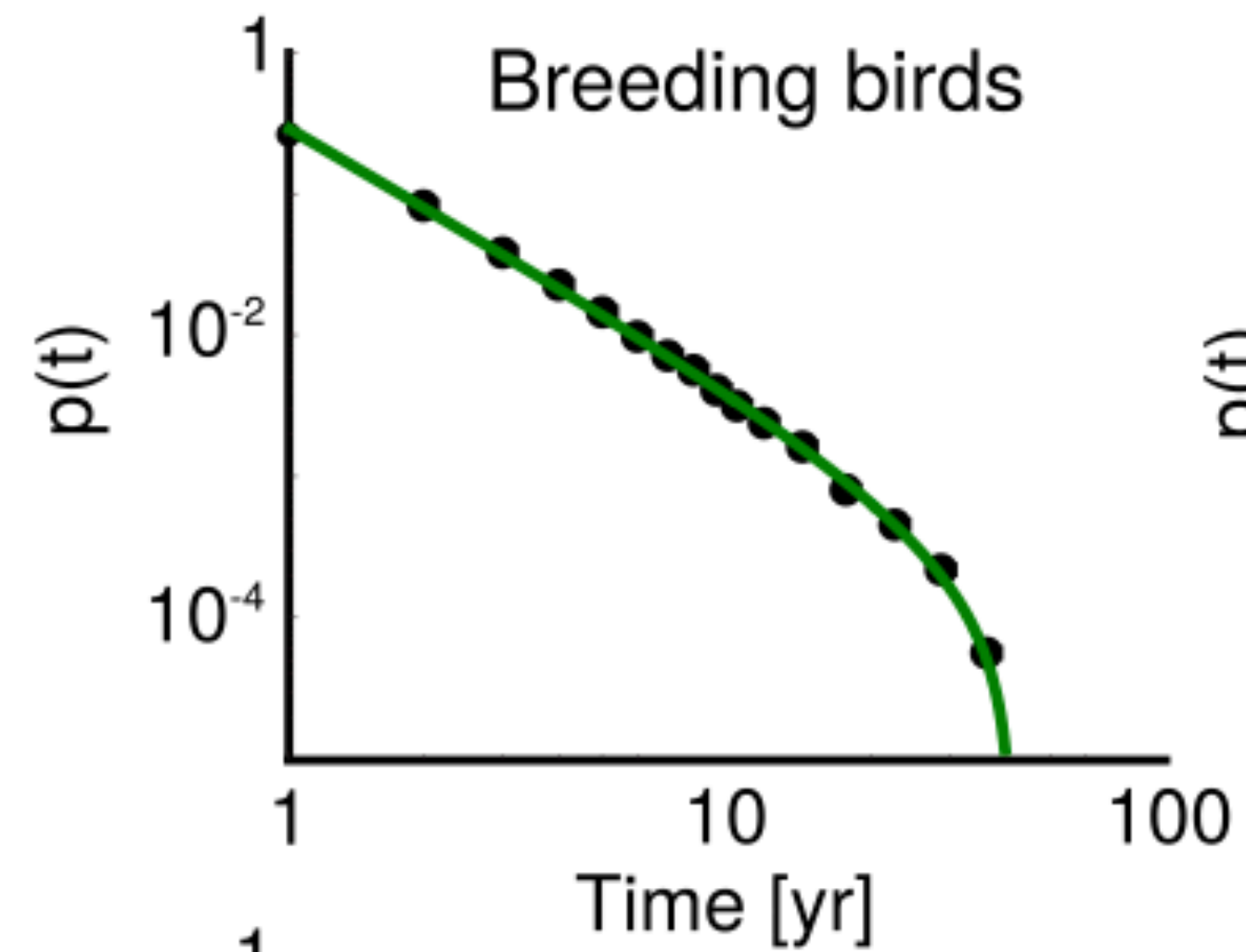
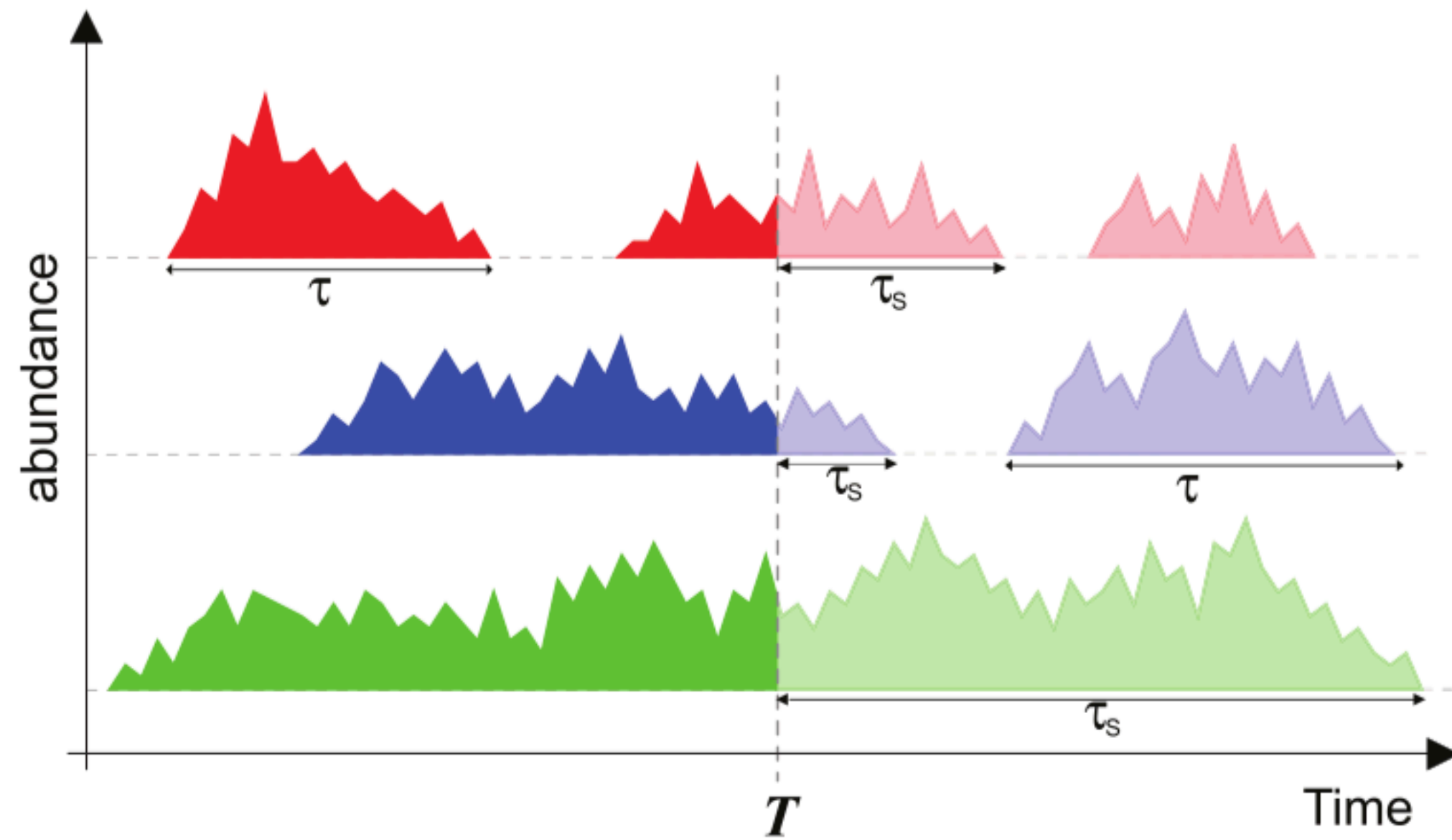
$$r = \log \lambda$$

$$g(r) = e^r P(e^r, t)$$



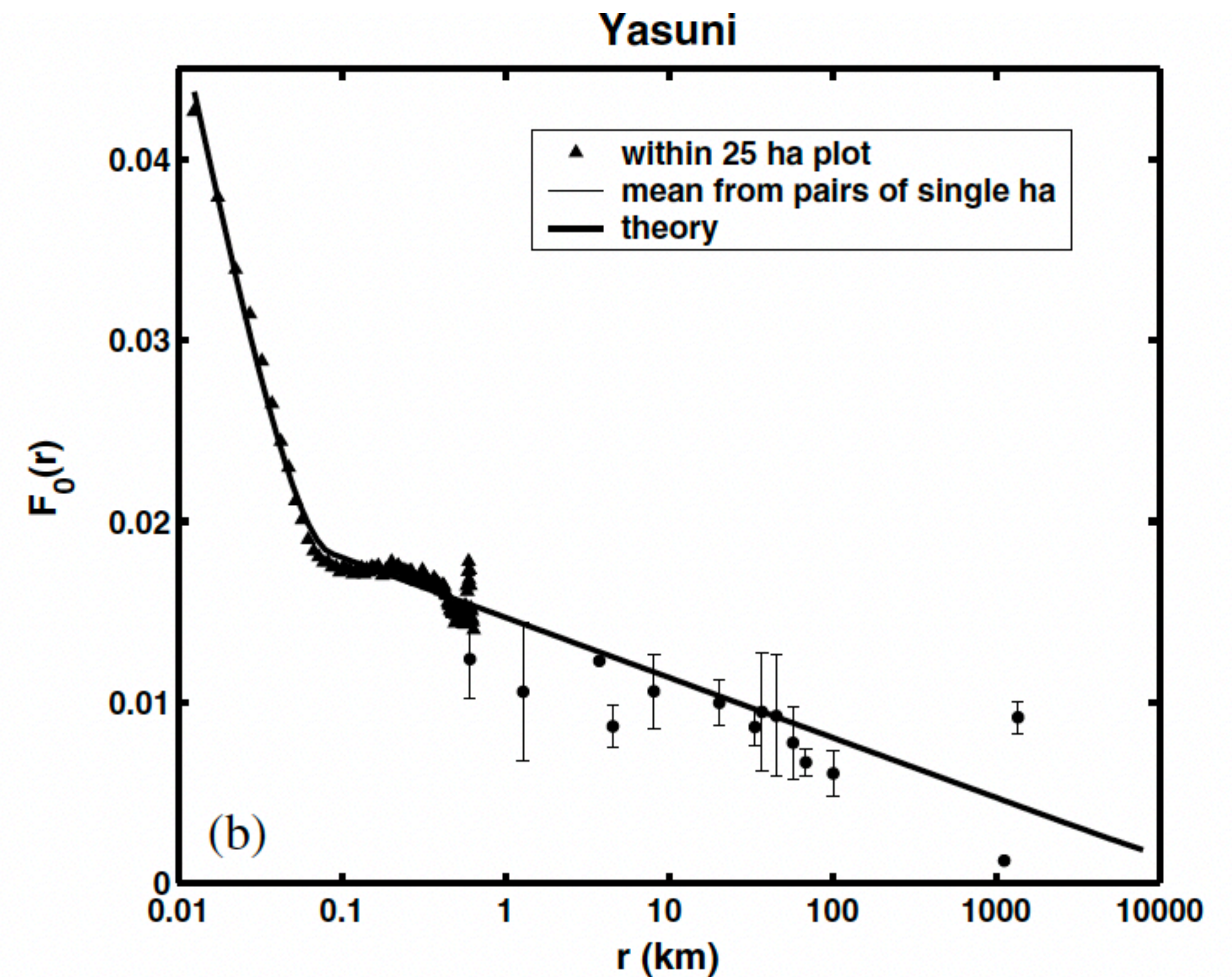
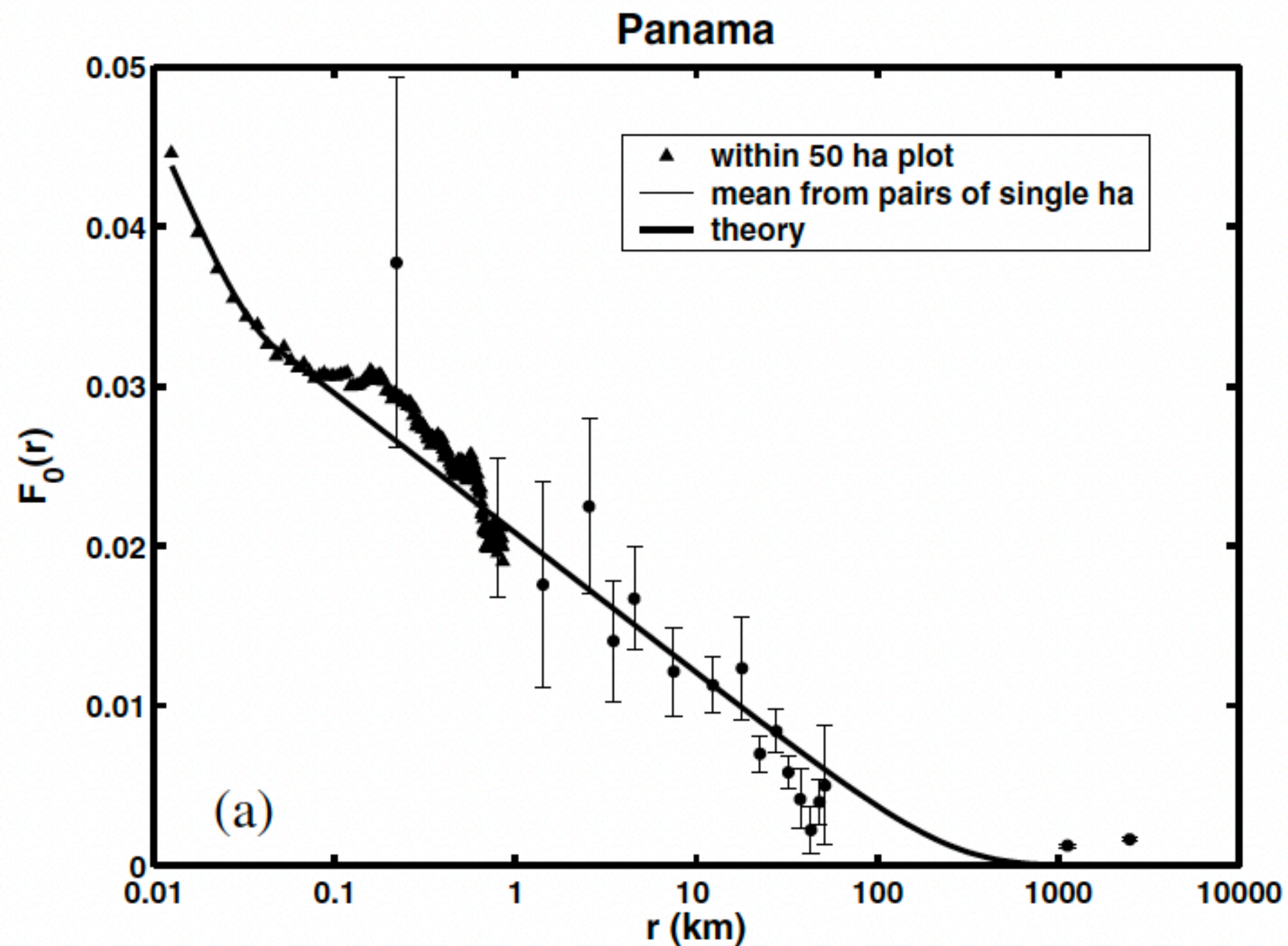
[Azaele et al., 2006]

Dynamics



Space

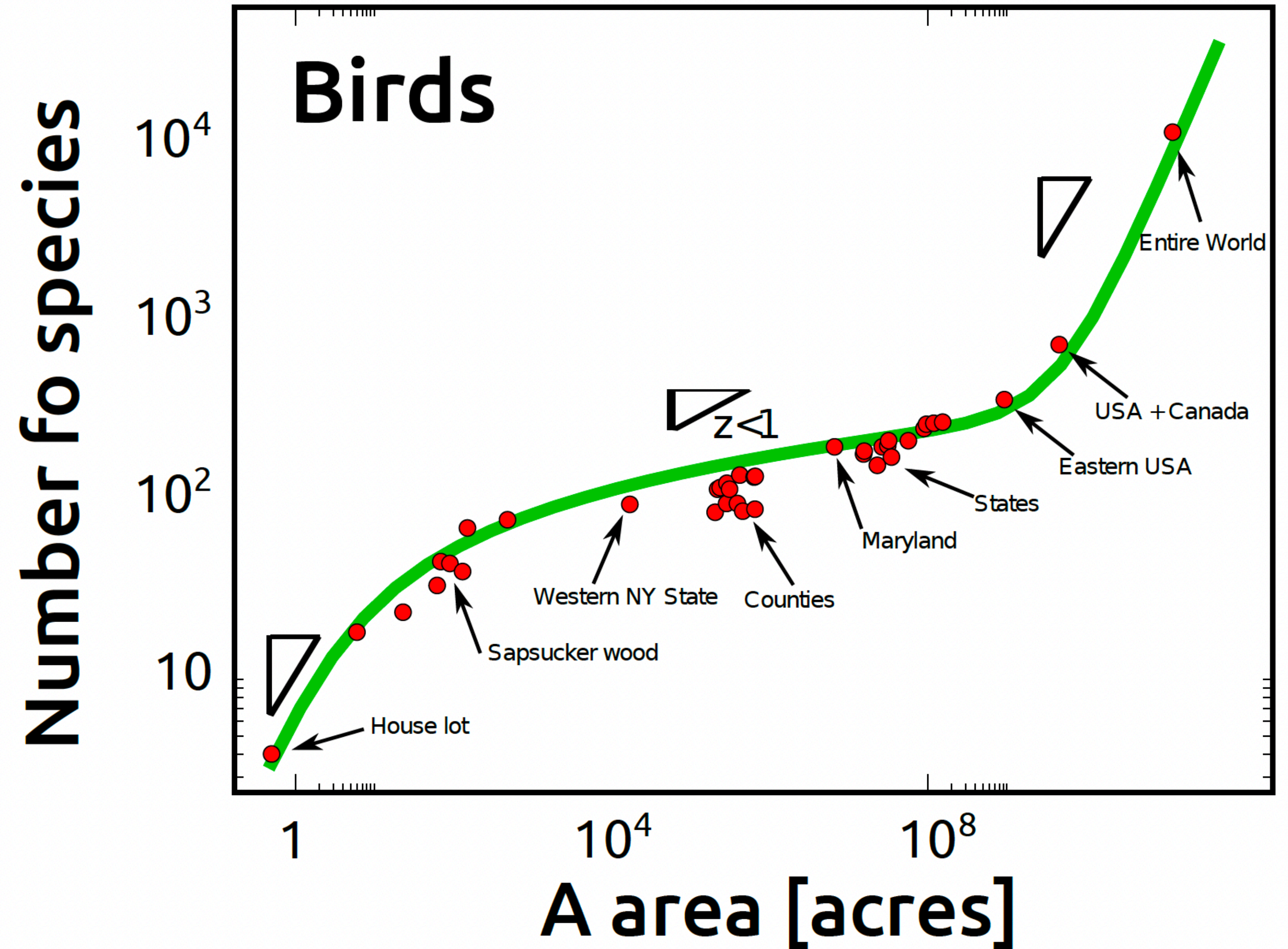
Probability that two random individuals are conspecific



[Zillio et al., 2015]

Space

Species Area Relationship



[Grilli et al., 2012]

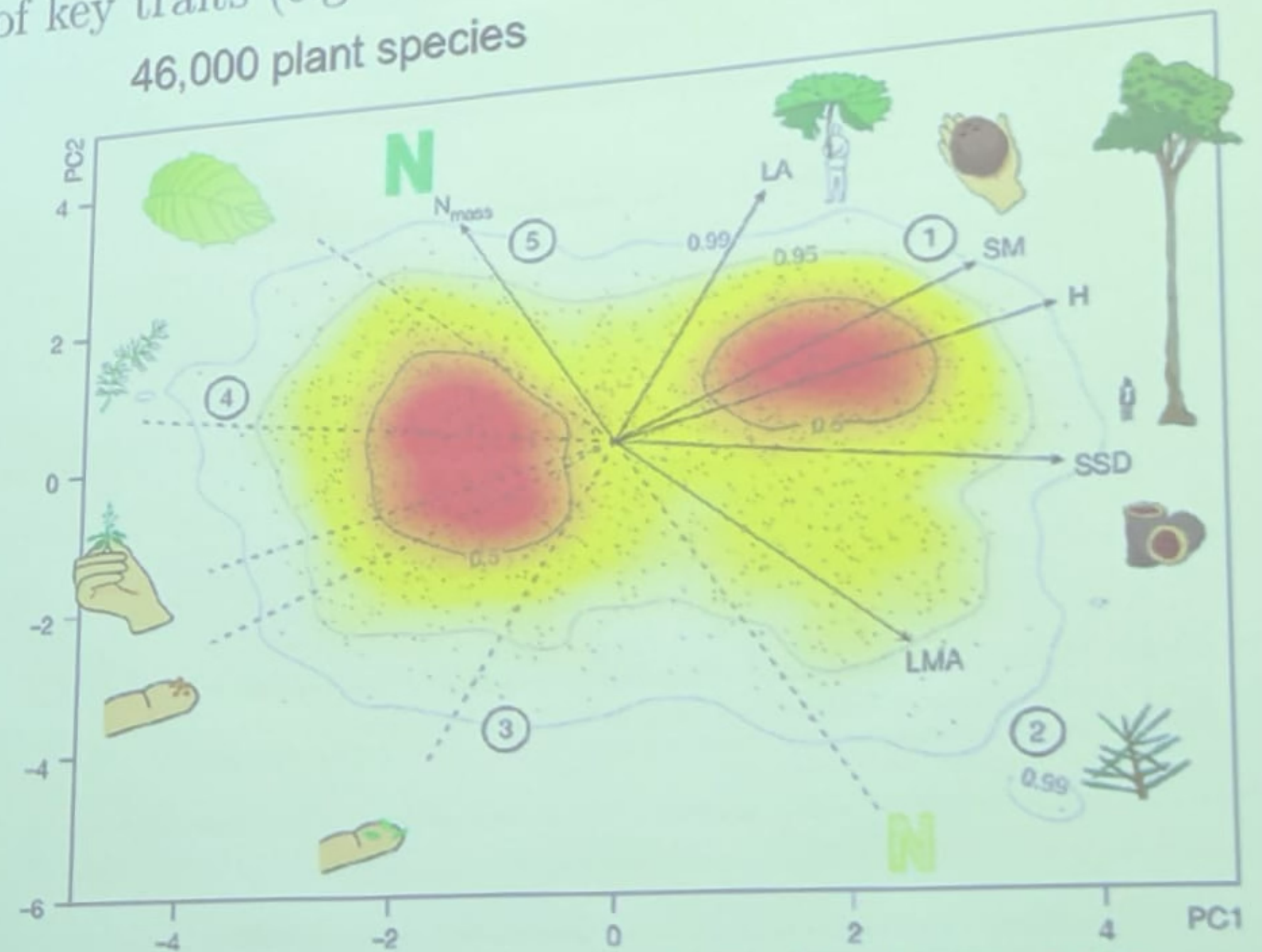
How neutral theory fails

Species are very different

Low-dim tradeoff surfaces

Like for environments, in each class of organisms, few axes (low surface) of key traits (e.g. for plants: leaf and seed features)

46,000 plant species



Diaz et al, Nature, 2016

Species differences play a role

Key result: Species that have stronger self-limitation in seedling recruitment are less common as adults in the forest.



Timescales

Speciation as source of diversity

How “old” is a species with a given relative abundance x ?

$$a(x) = -2N \frac{x}{1-x} \log x$$

[Kimura 1983]

Amazon rainforest

Total number of trees: $N \sim 10^{11}$

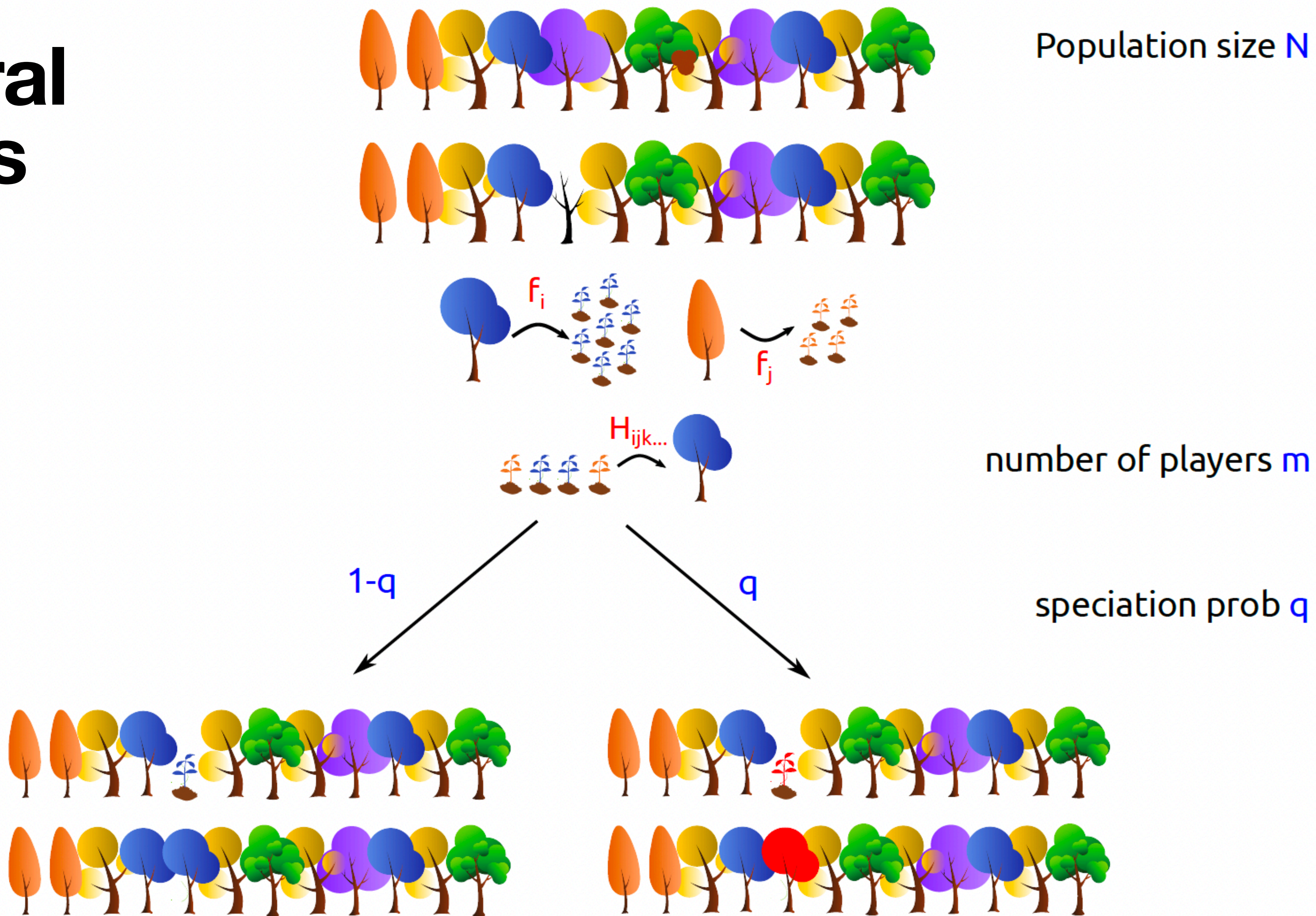
Most abundant species tree: $x \sim 10^{-2}$

Generation time of trees: ~ 30 years

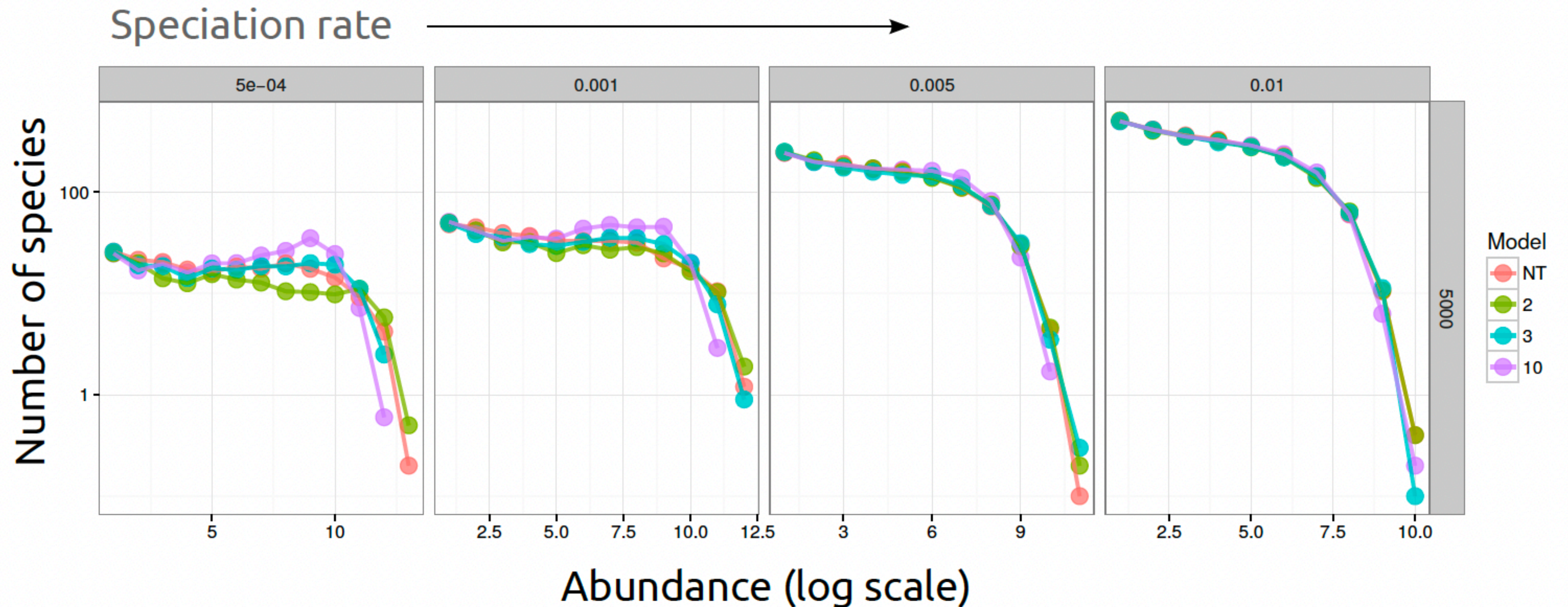
Age of the most abundant species: $\sim 10^{11}$ years

Age of the universe: $\sim 10^{10}$ years , First tree: $\sim 10^8$ years ago

Non-neutral extensions



Species Abundance Distribution is insensitive to interactions



[Grilli et al., 2017]

Time to extinction is sensitive to neutrality

